Newton School of Technology

2

Hardware & Software

Team Newton

C02: System & AI Essentials

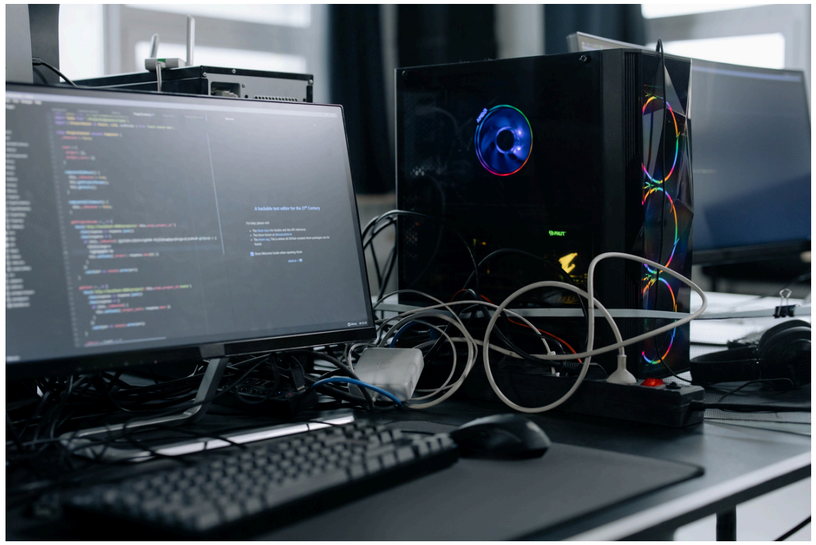
Newton School of Technology

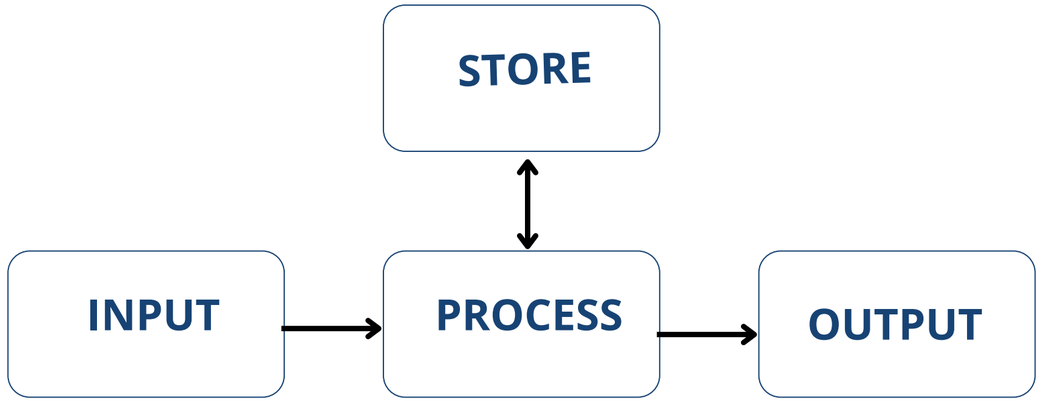
Join the lecture through the online portal

Recap

A computer is an electronic device that takes **input**, **processes** it, and gives **output**.

It follows a set of instructions (called a program) to perform tasks like calculations, writing, drawing, playing music, and more.





Newton School of Technology

What Makes a Computer Work?

Newton School of Technology

Hardware

Software

Hardware

Hardware is the **physical parts** of a computer — the things you can see and touch.

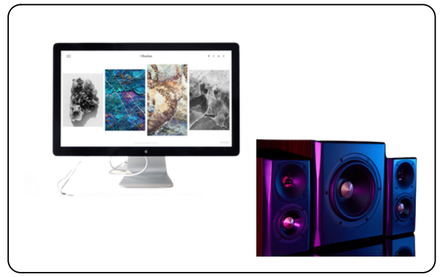
Examples:

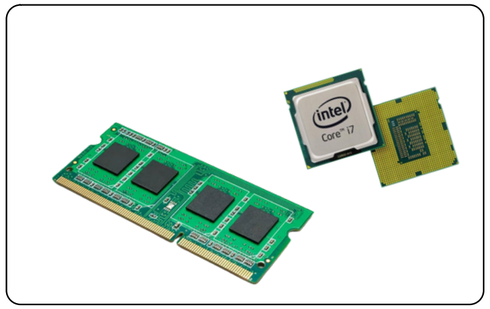
1. Monitor
2. Keyboard & Mouse
3. CPU
4. RAM
5. Hard Disk
6. Printer

Hardware = **The body of the computer**

Storage Devices

Input Devices

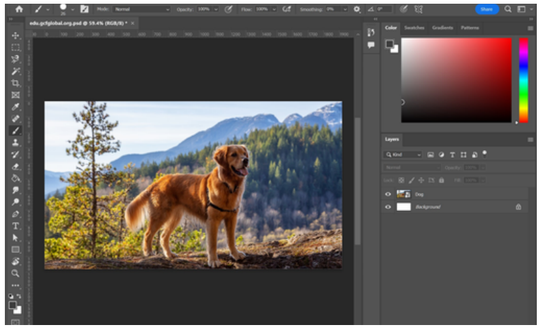

Output Devices



Compute Devices

Software

Software is a set of **instructions** or programs that tell the computer what to do and how to perform tasks.
Examples: MS Word, MS Excel, PowerPoint, Chrome, etc.



Software helps you:

- Browse the internet
- Play games
- Write documents
- Edit photos
- Watch videos
- Manage files

Let’s open up a Computer!

Input Devices

Input devices are tools we use to **send data or commands** to a computer.

Keyboard

Mouse

Mic

Digital Pen

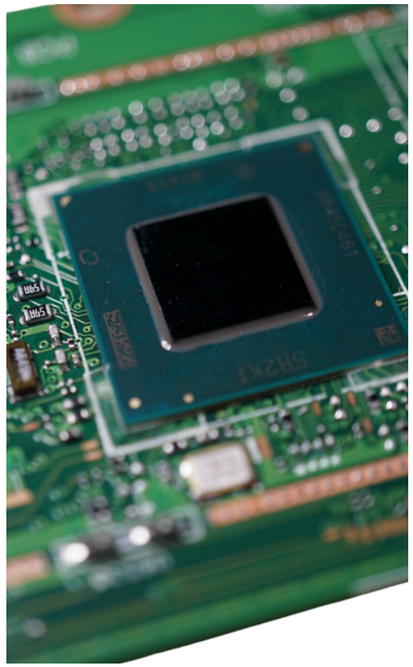
Webcam

Touchscreen

Scanner

Game controller

Internal Hardware



CPU

HDD

RAM

GPU

Motherboard

CPU (Central Processing Unit)

The **Central Processing Unit (CPU)** is like the brain of a computer. It's the part that does most of the **thinking**, **calculating**, and **decision-making** to make your computer work.
Whether you're playing a game, typing a school assignment, or watching a video, the CPU is busy handling all the instructions.

Front

Back

The CPU is usually placed in a special slot called a **socket** on the computer's **motherboard**, which is like the main circuit board that connects all the parts of a computer

GPU (Graphics Processing Unit)

A GPU, or Graphics Processing Unit, is a part of the computer that helps to create and display images, videos, and animations on your screen. It works alongside the CPU to make sure visuals are shown smoothly and quickly.

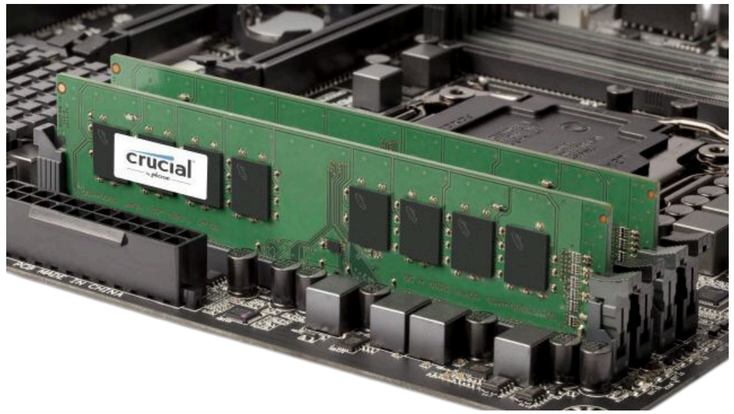
What Does a GPU Do?

- Helps in showing high-quality images and videos
- Makes games,editing videos, pictures and animations run smoothly
- Speeds up work in 3D modeling, virtual reality, and design software

RAM (Random Access Memory)

Newton School of Technology

RAM (Random Access Memory) is a computer's temporary, high-speed memory that **stores data and instructions currently being used by running programs**.



It's volatile, meaning its contents are cleared when the power is turned off, unlike permanent storage such as an SSD or hard drive.

Output Devices

Output devices are tools a computer uses to **show results or give feedback** to us.

Monitor

Headset without Mic

Speaker

Projector

Printer

Secondary Storage

Secondary storage is long-term storage that keeps data even when the computer is turned off, such as SSDs, hard drives, and USB drives.

1. Hard Disk Drive (HDD)

- Uses magnetic disks to store data.
- Has moving parts (like spinning platters and read/write heads).

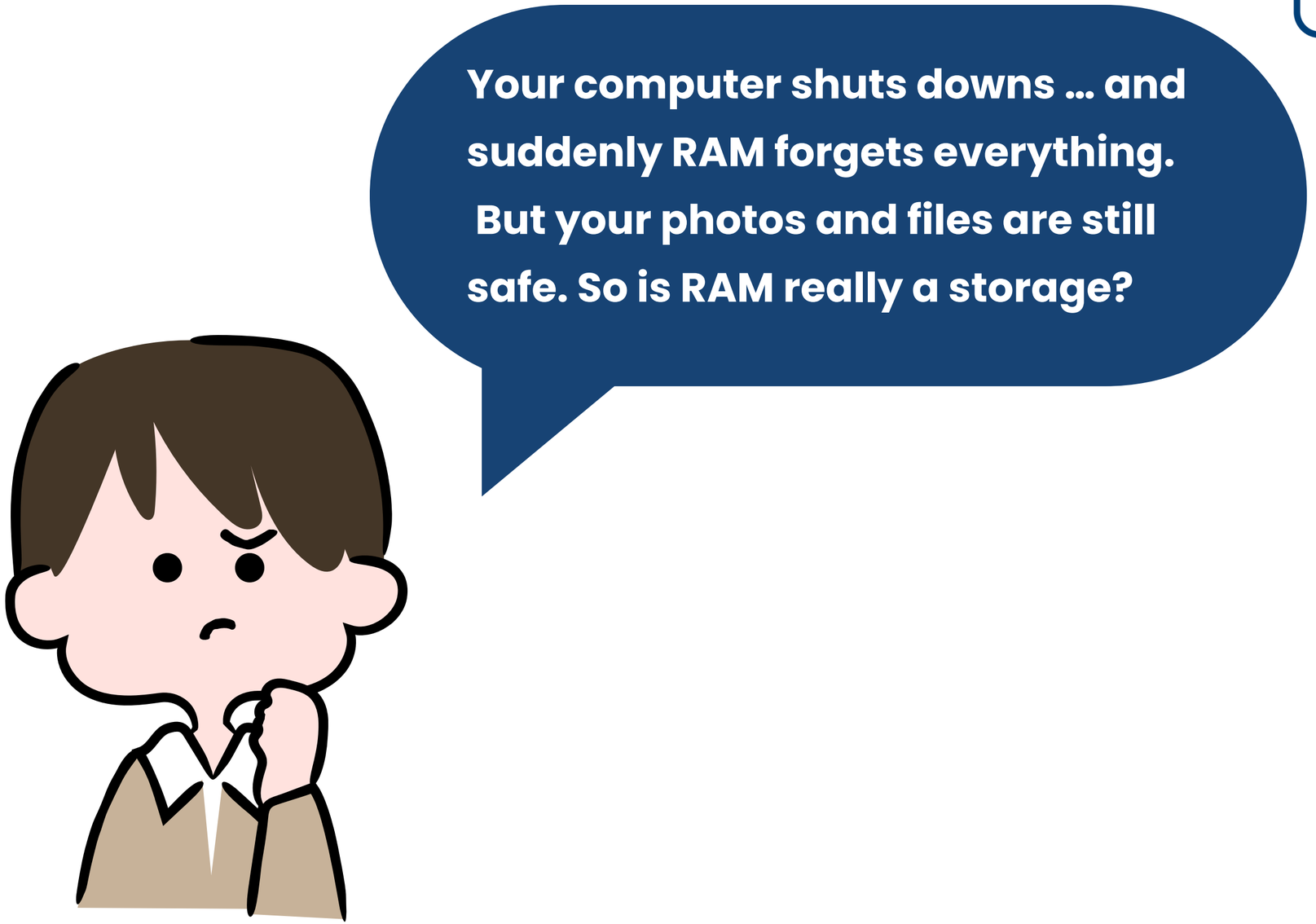
2. Solid State Drive (SSD)

- Uses flash memory (no moving parts).
- Much faster than HDD in reading and writing data.

Newton School of Technology

Hardware Activity

Newton School of Technology

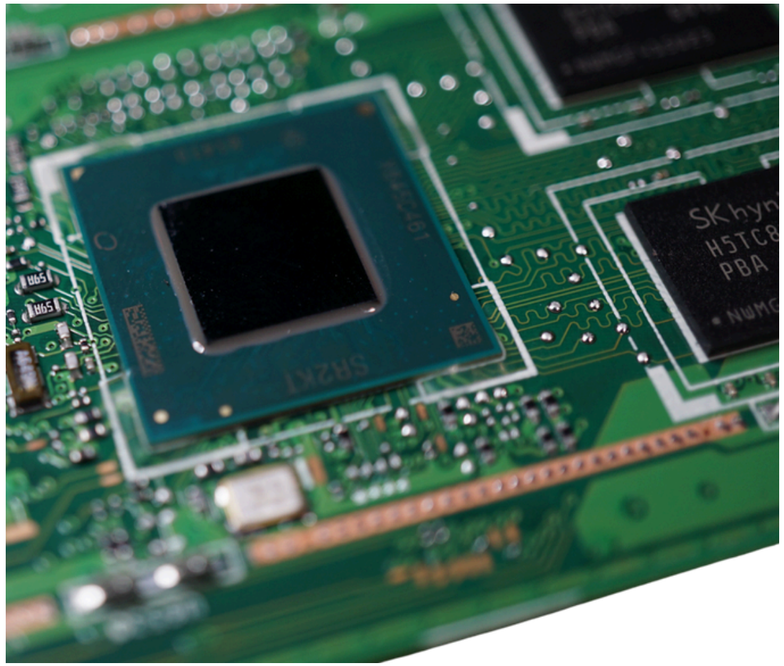


RAM : Is it storage or memory ?

- RAM is Memory, not permanent storage.
- It temporarily stores data and programs that the computer is currently using.
- Data in RAM is lost when the computer is turned off (volatile memory).
- RAM helps the system run applications faster by providing quick access to active data.

Newton School of Technology

CPU



GPU

Newton School of Technology

How GPU is different from CPU ?

CPU (Central Processing Unit)

- Brain of the computer
- Few powerful cores
- Best for general tasks and decision-making

GPU (Graphics Processing Unit)

- Specialized processor for graphics and parallel tasks
- Thousands of smaller cores
- Best for gaming, video editing, and AI

Newton School of Technology

Inclass Quiz

How Does Hardware Know What To Do?

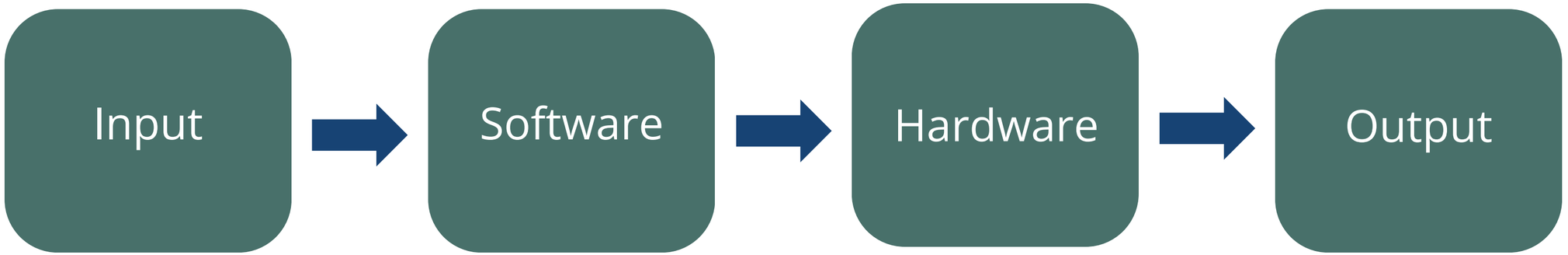
Hardware Alone Cannot Work

- CPU does not know which calculation to perform.
- Monitor does not know what to display.
- Printer does not know what to print.

Newton School of Technology

Software

Software Gives Instructions to Hardware



- Software tells hardware what tasks to perform.
- It controls how hardware behaves.
- Without software, hardware is useless.

What is Software ?

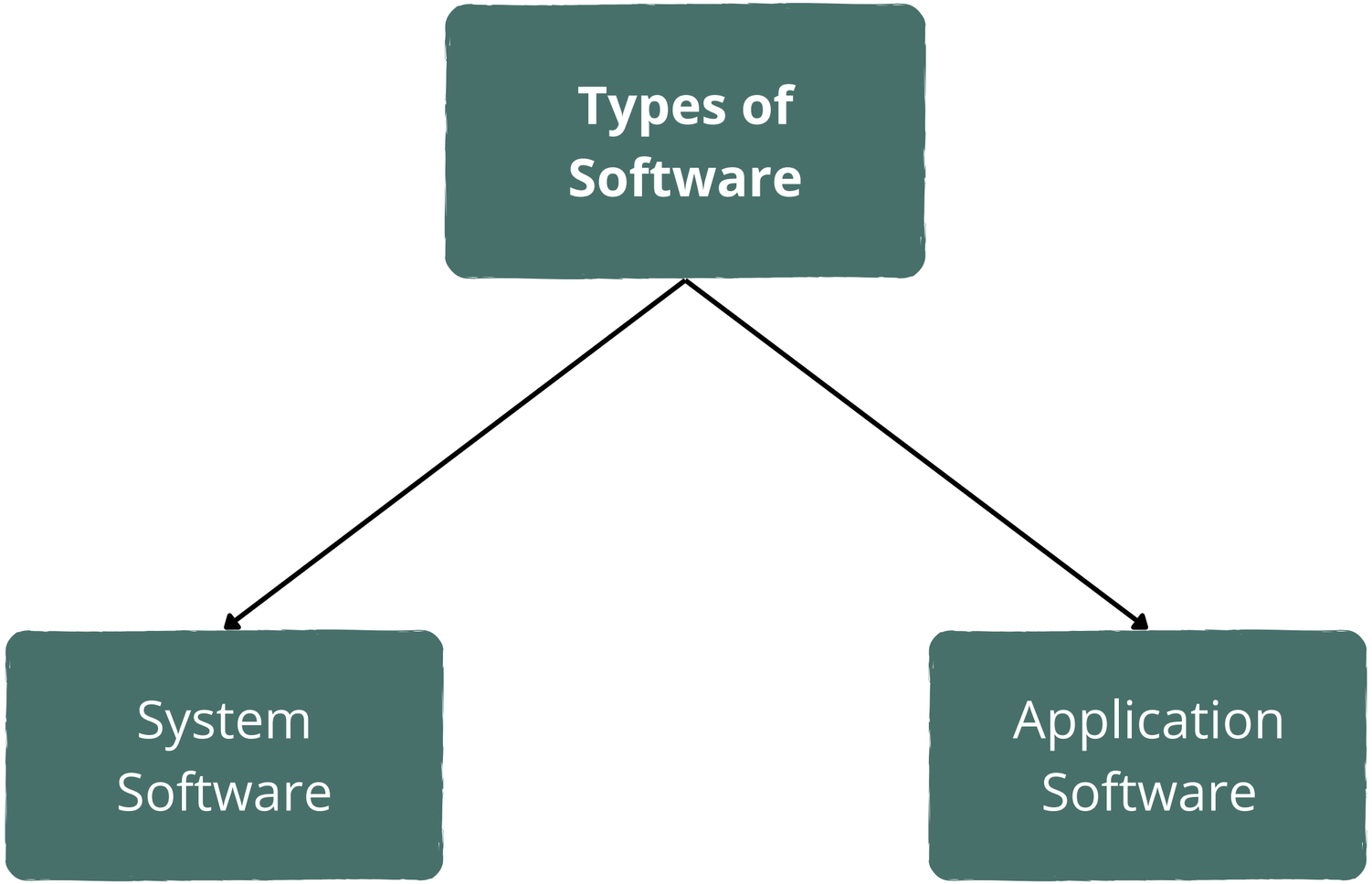
Software is a collection of programs and instructions that tell a computer what to do.

Example : Playing Music

- Software → Spotify/VLC
- Hardware → CPU, RAM, speakers

Newton School of Technology

Types of Software


Types of Software
System Software
Application Software

System Software

Newton School of Technology

System software is a type of computer **program** that controls and manages a computer's hardware and system resources—such as **memory**, **processors**, and devices—while providing a platform for applications and other programs to run.

Examples: Operating System, Device Drivers, Compilers.

Examples of System Software

Operating Systems

Programming Languages

Package Managers

Drivers

System Utilities

Application Software

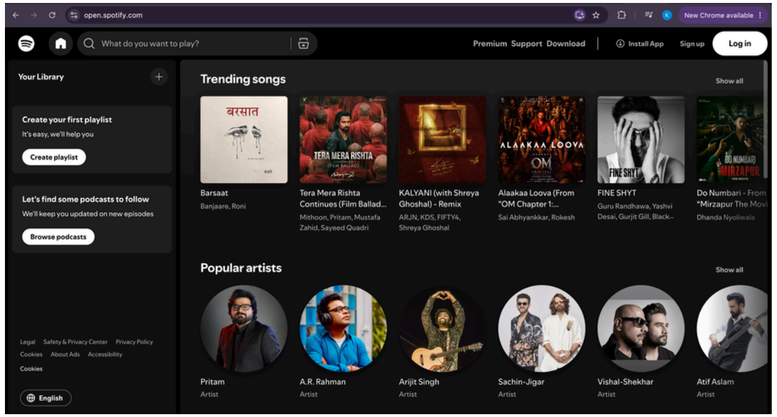
Newton School of Technology

Application software is a type of computer program that allows users to perform useful tasks and relies on **system software** to access and use the computer's **hardware**.

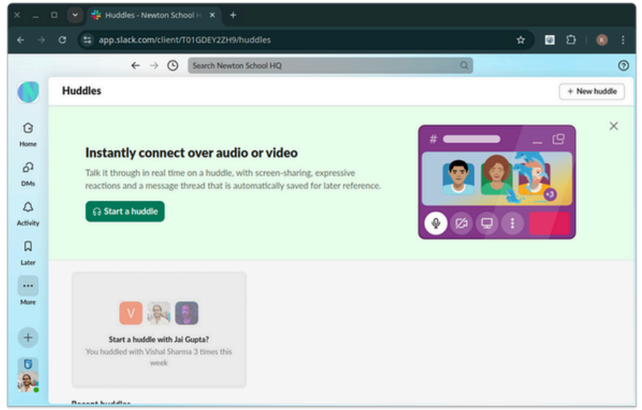
Examples: Media Players, Browsers, Databases, Games.

Examples of Application Software

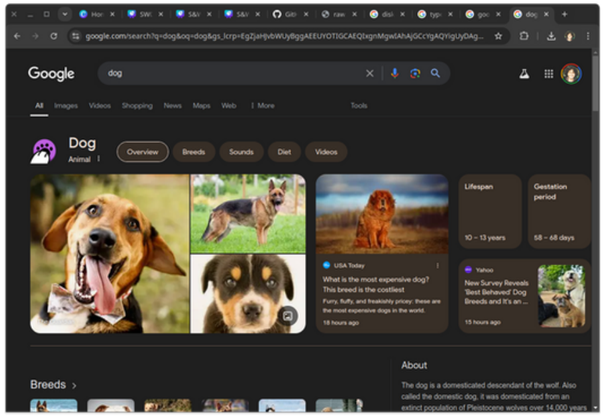
Games



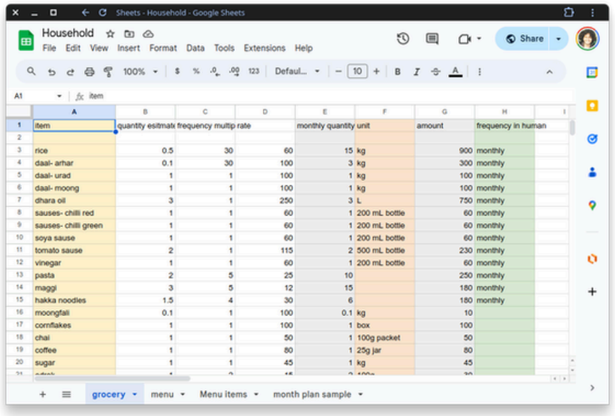
Spotify



Messaging Apps

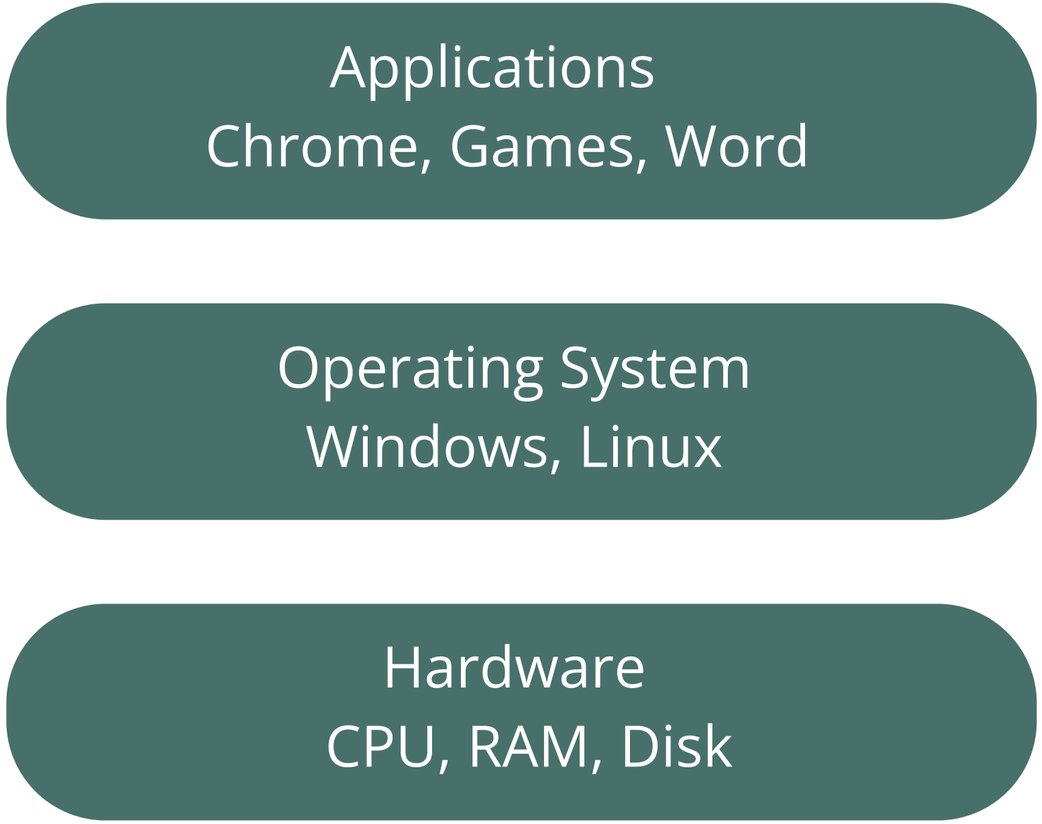


Browsers



Office Tools

The Software Stack



Newton School of Technology

Operating Systems

The OS: Face of the Machine

Newton School of Technology

When you switch on a computer, phone, or laptop, the first software you interact with is the Operating System (OS).
You do not directly interact with the **CPU, RAM, or storage**.

Example: A laptop running:

- Windows feels different from
- macOS, which feels different from
- Linux

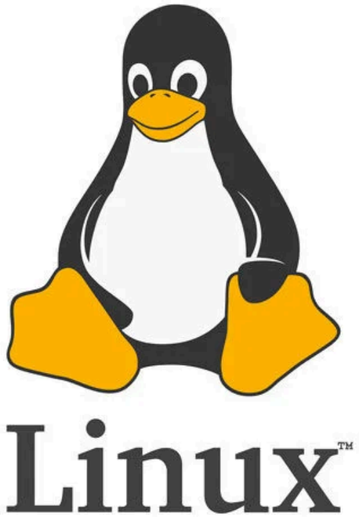
Even if the CPU, RAM, and storage are similar.

macOS

The operating system running on MacBooks, designed for tight integration with Apple hardware and a clean user experience.

GNU/Linux

An open-source operating system used on laptops, servers, cloud systems, and supercomputers around the world.



Microsoft Windows

The most widely used desktop operating system globally, common in homes, schools, and enterprise environments.

Android

A mobile operating system designed for smartphones and tablets, focused on touch interaction, portability, and battery efficiency.

Key Differences

Feature	Windows	Mac	Linux
Ease of Use & Learning Curve	User-friendly for all drivers, low to moderate learning curve	Extremely user-friendly, very low learning curve	Requires technical knowledge, moderate to high learning curve
Customization & Flexibility	Limited but accessible	Minimal, controlled by Apple	Highly customizable, DIY-friendly
Software & Hardware Compatibility	Wide range of applications, broad hardware support	Limited to the Apple ecosystem	Extensive range but requires tuning, broad hardware support
Security & Maintenance	Moderate security, regular updates	High security, minimal maintenance	High security, frequent tune-ups, user-managed
Target Users	Everyday users, business professionals	Professionals, creatives	Tech enthusiasts, developers

Same Hardware, Different Experience

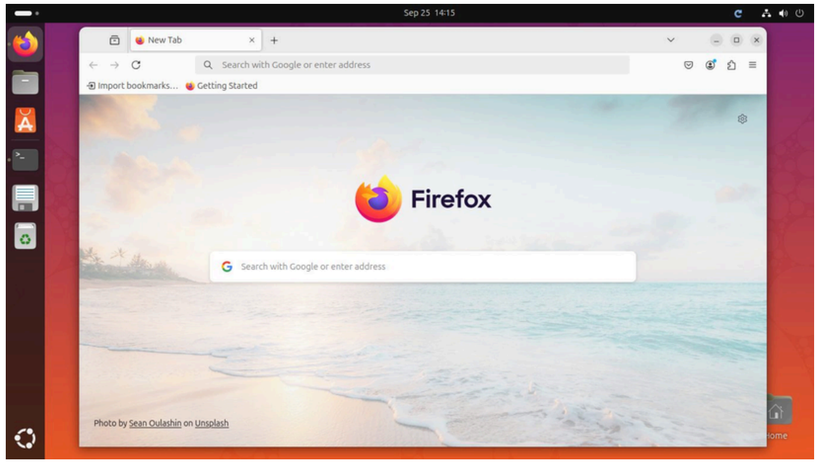
Newton School of Technology

Today we will look at the same machine running a different Operating System: Ubuntu Linux.

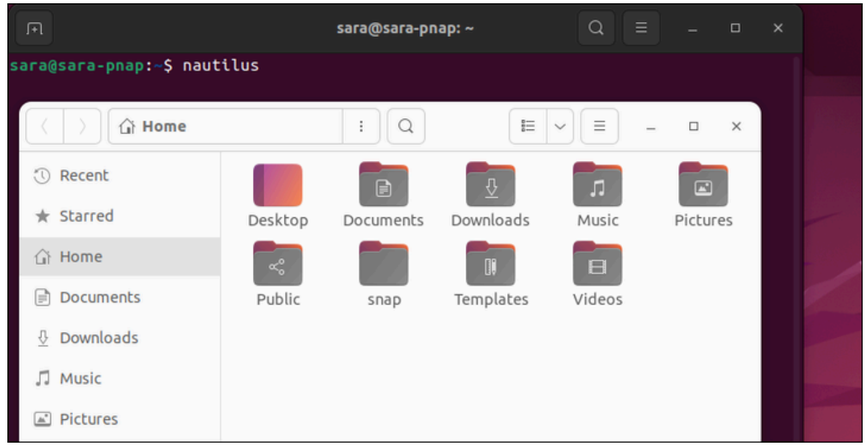
We will explore:

- File Manager
- Web Browser
- Terminal

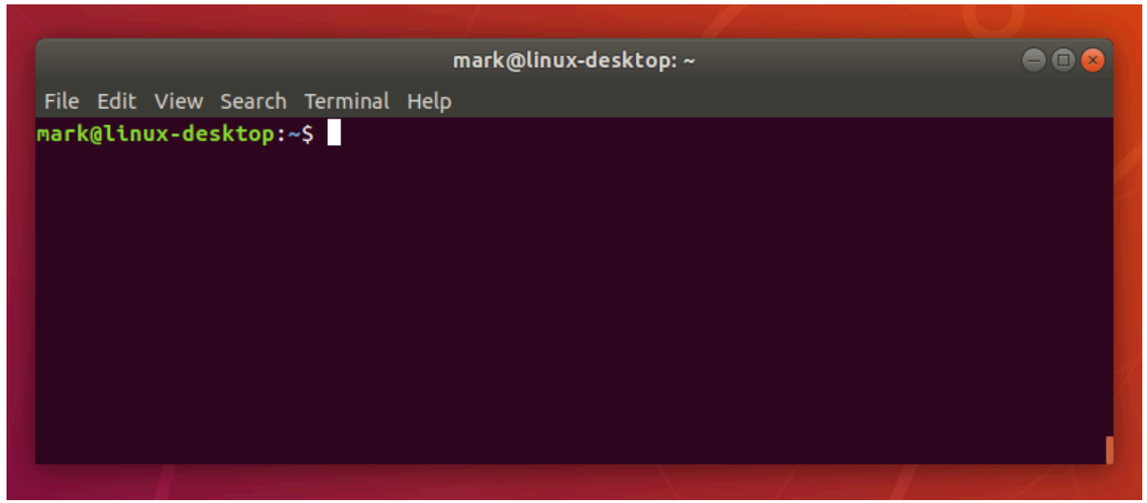
Even though the hardware is unchanged, the system looks and behaves differently because the OS has changed.



Browser

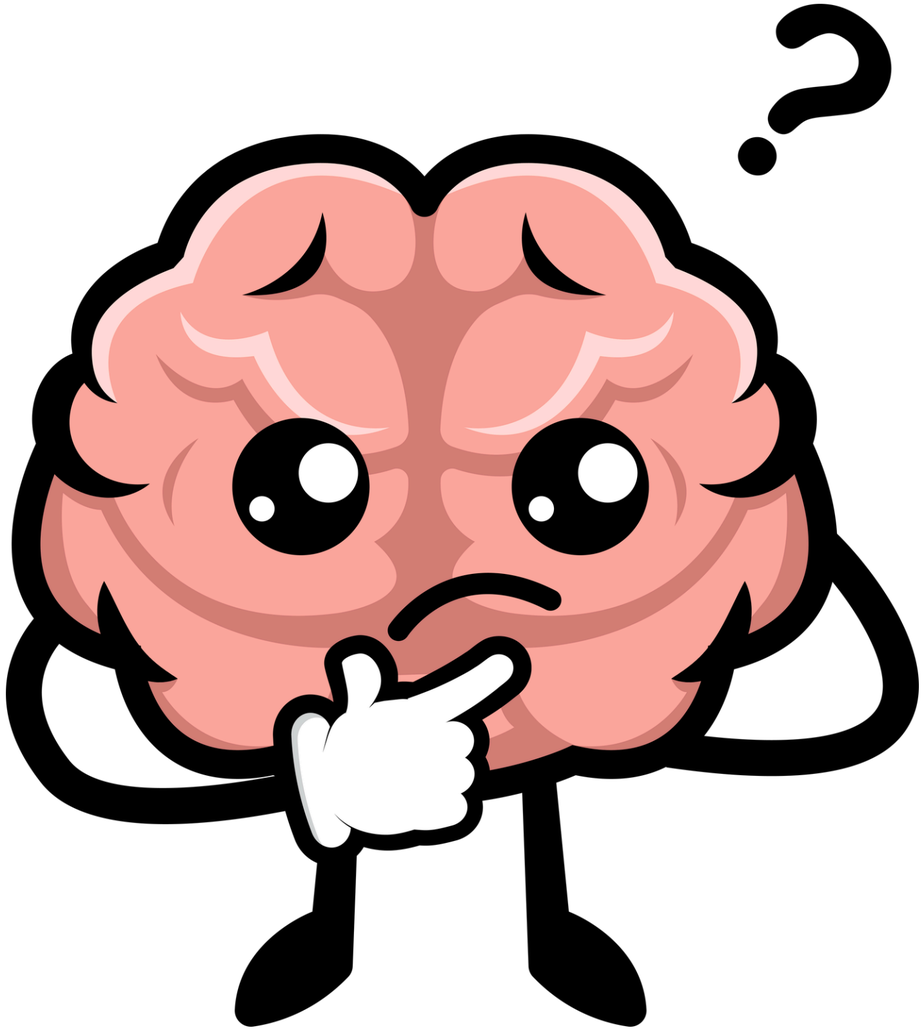


File Manager



Terminal

What is familiar ? What is different ?



What is familiar ? What is different ?

Newton School of Technology

Familiar Things

- Desktop, windows, icons, and taskbar/dock
- File manager with folders and files
- Browser works the same way (tabs, search, downloads)

Different Things

- Terminal is used much more often and is more central
- Ubuntu interface/design looks different from macOS and Windows
- Software is often installed through package managers instead of app store

Newton School of Technology

Inclass Quiz

Further Reading

- https://byjus.com/govt-exams/input-and-output-devices/
- https://www.ibm.com/think/topics/hardware
- https://www.geeksforgeeks.org/computer-science-fundamentals/software-and-its-types/
- https://www.geeksforgeeks.org/computer-science-fundamentals/what-is-a-storage-device-definition-types-examples/
- https://www.almabetter.com/bytes/articles/what-is-memory-hierarchy
- https://www.intel.com/content/www/us/en/products/docs/processors/cpu-vs-gpu.html
- https://www.geeksforgeeks.org/operating-systems/types-of-operating-systems/
- https://en.wikipedia.org/wiki/Ubuntu

Feedback

Now it's time to give your valuable feedback

Newton School of Technology

Thank You!